Executive Summary

1. Project Overview

This project analyzes the **Telco Customer Churn dataset** to identify the major factors influencing customer attrition and retention. The dataset contains information on customer demographics, account details, contract types, and subscribed services. The primary objective of the analysis is to uncover patterns associated with customer churn and provide actionable recommendations that can help improve customer retention.

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2. Overall Churn Analysis

The analysis revealed that out of **7,043 total customers**, **1,869 customers (26.5%) churned**, while **5,174 customers (73.5%) remained with the company**. Although the majority of customers were retained, the churn rate is significant enough to impact revenue and customer growth, highlighting the need for effective retention strategies.

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3. Demographic Insights

Gender Analysis

The visualizations indicate that **gender has little to no influence on customer churn**. Male and female customers exhibit nearly identical churn behavior, suggesting that retention efforts should focus on other customer characteristics rather than gender-based segmentation.

Senior Citizen Analysis

The analysis shows that **senior citizens experience a higher churn rate than non-senior customers**. This indicates that older customers may face challenges related to pricing, service complexity, or support needs and could benefit from targeted customer engagement programs.

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4. Customer Tenure Analysis

Customer tenure emerged as one of the strongest predictors of churn. The visualizations demonstrate that:

- Customers with **short tenure periods are more likely to churn**.
- Customers who have remained with the company for a longer period exhibit significantly lower churn rates.

- Long-term customers appear more satisfied and loyal, indicating that retention efforts should focus heavily on the early stages of the customer lifecycle.

These findings suggest that improving customer onboarding and engagement during the first few months could significantly reduce churn.

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5. Contract Type Analysis

Contract type was identified as one of the most influential factors affecting customer retention.

Key Findings:

- Customers with **Month-to-Month contracts show the highest churn rates**.
- Customers on **One-Year contracts exhibit considerably lower churn**.
- Customers on **Two-Year contracts have the lowest churn rates**, indicating the strongest customer loyalty.

This pattern demonstrates that longer-term contractual commitments significantly improve customer retention and reduce the likelihood of customer attrition.

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6. Internet Service Analysis

Internet Service Type

The analysis of internet services revealed notable differences in churn behavior:

- **Fiber Optic customers contribute the largest share of churned customers.**
- DSL customers display comparatively lower churn rates.
- Customers without internet services show the lowest churn levels.

The unexpectedly high churn among Fiber Optic customers may indicate issues related to pricing, customer expectations, service quality, or competition.

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7. Service Subscription Analysis

Several value-added services show a strong relationship with customer retention.

Online Security

Customers without Online Security services show substantially higher churn rates compared to customers who subscribe to the service.

Online Backup

Customers who utilize Online Backup services are more likely to remain with the company, while customers without this service exhibit increased churn.

Device Protection

The presence of Device Protection is associated with lower churn levels, indicating that customers who invest in additional service protection tend to be more engaged.

Tech Support

Customers without Tech Support services experience one of the highest churn rates among service categories, highlighting the importance of customer assistance and support programs.

Streaming Services

Streaming TV and Streaming Movies services show moderate influence on churn. While churn exists across both groups, the impact is not as significant as security and support-related services.

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8. Phone Service and Multiple Lines Analysis

The analysis indicates that:

- Phone Service alone has a relatively weak relationship with churn.
- Customers with Multiple Lines show similar churn behavior to customers without Multiple Lines.
- These features are less influential compared to contract type, tenure, and internet-related services.

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9. Key Drivers of Customer Churn

Based on the exploratory analysis and visualizations, the primary drivers of churn are:

1. **Month-to-Month Contracts**
2. **Short Customer Tenure**

3. **Fiber Optic Internet Service**
4. **Lack of Online Security**
5. **Lack of Tech Support**
6. **Lack of Online Backup**
7. **Lack of Device Protection**
8. **Senior Citizen Status**

These factors consistently appear across multiple visualizations as strong indicators of customer attrition.

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10. Business Recommendations

Based on the findings, the following strategies are recommended:

Improve Contract Retention

Encourage Month-to-Month customers to transition to One-Year or Two-Year contracts through discounts and loyalty incentives.

Enhance Customer Onboarding

Implement proactive engagement programs for new customers during their first year to reduce early churn.

Strengthen Support Services

Promote Online Security, Online Backup, Device Protection, and Tech Support through bundled packages and awareness campaigns.

Investigate Fiber Optic Churn

Conduct additional research to identify the reasons behind the higher churn rate among Fiber Optic customers and improve service quality where necessary.

Focus on High-Risk Segments

Develop specialized retention strategies for senior citizens and customers with short tenure periods.

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11. Conclusion

The analysis demonstrates that customer churn is primarily driven by **contract type, tenure, internet service category, and the adoption of value-added support services**. While demographic factors such as gender have minimal influence, customers with short tenure, Month-to-Month contracts, Fiber Optic services, and limited service subscriptions exhibit significantly higher churn rates. By focusing on these high-risk customer segments and enhancing customer engagement initiatives, the company can reduce churn, improve customer satisfaction, and strengthen long-term profitability.